Haoyan Luo

Ph.D. Student in Computer Science | Agent & Interpretability
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Education

University of Cambridge — Cambridge, UK
Ph.D. in Computer Science; Supervisor: Mateja Jamnik — *May 2025 - Present*

- **Research topics**: Interpretability and Control of Large Language Models

Imperial College London — London, UK
M.Res. in AI & Machine Learning; Supervisor: Lucia Specia — *Sep. 2023 - Sep. 2024*

- **Thesis**: Explainability in Large Language Models: Pathways to Refinement and Alignment
- **Award**: Endowment Fellowship; Distinction Honours

The Chinese University of Hong Kong, Shenzhen — Hong Kong / Shenzhen, China
B.Eng. in Computer Science; First-Class Honours — *Sep. 2019 - July 2023*

- **Honors**: Dean's List of School of Data Science (AY21-23); Undergraduate Research Awards (AY21-22); Undergraduate Student Teaching Fellow Award (AY21-22); Shaw College Outstanding Graduates Award
- **Services**: President of **TED***xCUHK(SZ)*; Student Coordinator, Shaw College

Experience

Singapore AI Safety Hub — Singapore
AI Safety Research Fellow — *Sep. 2026 - Present*

- **Agent Monitoring** : Developing stronger monitor models to detect unsafe behaviour in multi-agent systems.
- **Scalable Interpretability** : Verbalising LLMs' hidden reasoning in natural language.

Vectify AI — London, UK
Research Intern (Part-Time) — *April 2025 - Nov. 2025*

- **PageIndex** : Built a reasoning-based RAG system that simulates how human experts navigate and extract knowledge from long documents through tree search; Github 30K+ stars. [code]

J.P. Morgan — London, UK
AI & NLP Associate — *Sep. 2024 - March 2025*

- **TIDE - Topic Insigts and Document Exploration** : Led the development of topic modeling and topic distillation pipelines in TIDE, our internal state-of-the-art solution powered by LLMs.
- **Agentic Hallucination Detection** : Developed unified hallucination detection pipelines tailored for financial agents.

Shanghai AI Laboratory — Shanghai, China
Research Intern, supervised by Dr. Xun Zhao — *July 2023 - Sep. 2023*

- **Mechanistic Interpretability**: Utilise the LM head to discover that the hidden states of transformers can be viewed as large memory blocks, with each of its cells being highly decomposable and explainable;

Microsoft Research Asia — Beijing, China
Research Intern, supervised by Dr. Jiang Bian & Dr. Wei Cao — *July 2022 - Nov. 2022*

- **Feature engineering**: Co-proposed OpenFE, a novel framework with feature boosting method to accurately identify useful new features for tabular data; Github 1K+ stars. [code]

Deep Reinforcement Learning Group — Shenzhen, China
Student Researcher, supervised by Prof. Shuang Li — *May 2022 – Nov. 2022*

- **Meta-policy on temporal point process**: Researched multivariate temporal point process and formulate the traffic congestion and epidemic (COVID-19) curbing problems as finite-time horizon RL problems. Embeded temporal point process dynamics into a meta-policy learning framework and extract explainable and transferrable information. [code]

Experience (continued)

Tencent Shenzhen, China

Big Data Engineer Intern *June 2021 – Sep. 2021*

- **Recommendation system**: Built a recommendation system sub-module with a session-based recurrent neural network and deployed it within the "news" page and "Daily Q&A" page in WeChat Mini Program;
- **Data warehouse construction**: Built Hive and ClickHouse support and used Scala (together with SQL insertion and Spark & Hadoop embedded features) to query information from the warehouse for data analysis;

Seasun Entertainment, Kingsoft Zhuhai, China

Software Engineer Intern *June 2020 – August 2020*

- **Backend development**: Built and maintained a large-scale intranet used to monitor users' behaviors and dialogue contents in a popular web game in China, JX3, with over 5 million active users;

Publications

Conference Papers

1. **[NeurIPS 2026 Spotlight] Haoyan Luo**, Mateo Espinosa Zarlenga, Mateja Jamnik. **Don't Lose Focus: Activation Steering via Key-Orthogonal Projections**, *The Fortieth Annual Conference on Neural Information Processing Systems*.
2. **[ICML 2023]** Tianping Zhang, Zheyu Zhang, Zhiyuan Fan, **Haoyan Luo**, Fengyuan Liu, Wei Cao, Jian Li. **OpenFE: Automated Feature Generation with Expert-level Performance**, *The Fortieth International Conference on Machine Learning*.

Preprints / Workshop Papers

1. **[InterpScience @ NeurIPS 2026] Haoyan Luo**, Mateo Espinosa Zarlenga, Mateja Jamnik. **Persistent Sparse Autoencoders: Learning Feature-Specific Timescales in Language Model Representations**, *Workshop on Interpretability as a Science, The Fortieth Annual Conference on Neural Information Processing Systems*.
2. **[AIW @ COLM 2026] Haoyan Luo**, Lucia Specia. **Tuning Language Models by Mixture-of-Depths Ensemble**, *Workshop on Actionable Interpretability, The Third Conference on Language Modeling*.
3. **[arXiv 2026]** Sae Young Moon, Myeong-jun Erik Jang, **Haoyan Luo**, Chunyang Xiao, Antonios Georgiadis, Fran Silavong. **TIDE - Industry-aligned Granular Topic Modeling**, *arXiv:2601.11762*.
4. **[arXiv 2024] Haoyan Luo**, Lucia Specia. **From Understanding to Utilization: A Survey on Explainability for Large Language Models**, *arXiv:2401.12874*.

Competitions and Awards

- Endowment Fellowship, Imperial College London 2023
- Meritorious Winner, Mathematical Contest in Modeling (MCM'22) 2022
- First Prize in Contemporary Undergraduate Mathematical Contest in Modeling, China (CUMCM'21) 2021
- Third Prize, RoboMaster University AI Challenge 2021
- National Silver Award, First Prize Award in Guangdong, Google App Inventor Challenge 2019

Service & Teaching

- **Reviewer:** EMNLP; ICML; ICLR; NeurIPS.
- **Teaching Assistant:** CUHK: CSC3170 (*Database Systems*) and DDA4020 (*Advanced Machine Learning*); University of Cambridge: *Machine Learning for Real-World Data*.

Technical Skills

Programming: Python, C++, PyTorch, Rust, MATLAB, Scala

Languages: Mandarin Chinese (Native), Cantonese (Fluent), English (Professional)